Reference-dependent instability in spatial agricultural suitability assessment: Evidence from rice cultivation-system transfer in Europe

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Abstract

Climatic thresholds transferred from a reference region are widely used to screen where a cropping system could be introduced, yet the reliability of the resulting maps is rarely assessed against mapped existing cultivation. I assess a growing-degree-day (GDD10) gate for transferring the cold-region rice system of Hokkaido, Japan, to 69.7 Mha of incumbent arable land in Europe (excluding Russia), using WorldClim 2.1, MapSPAM 2010 and the Global Map of Irrigation Areas on one 5′ grid. Calibrated on Hokkaido rice area, the reference is 589 GDD10; 92.1% of incumbent arable land passes it, but only 16.0% also has mapped irrigation infrastructure. In a controlled contrast for Germany, the pass share falls from 96.6% to 3.2% when a point-derived reference (~900 GDD10) replaces the grid-calibrated one, whereas recalibrating at 2.5′–10′ changes it by at most 2.4 percentage points. A threshold sweep (0–1800 GDD10) shows that the 501.1 thousand ha of mapped European rice harvested area is a weak arbiter between these references: raising the threshold from 589 to 900 removes 37.2 percentage points of candidate area but increases omitted rice area only from 1.37% to 3.22%. Omissions fall mainly in low-density allocation cells at climatic margins. In France, the gate omits 12.1% of

MapSPAM rice area at T^* but none of the 13.7–19.6 thousand ha declared in the national parcel register (2010–2024), which occupies only 29–39 warm cells and cannot rank references. Presence-only coverage constrains only implausibly strict thresholds; reference construction, not grid support, is the dominant, unvalidated source of uncertainty.

Keywords growing degree days · threshold sensitivity · omission analysis · MapSPAM · irrigation infrastructure · classification stability

1. Introduction

Spatial suitability assessment is a routine instrument of environmental and agricultural planning. Typical studies combine climatic, edaphic and topographic layers by weighted overlay, fuzzy membership or species-distribution modelling to delimit zones in which a crop could be grown (Azadi et al., 2023; Bazkiaee et al., 2024; Mohamed et al., 2019; Van et al., 2022). The same logic is applied at continental and global extents, where gridded climate normals and crop maps are screened against crop-specific requirements (Ojara et al., 2021; Zabel et al., 2014) and used to discuss crop migration under warming (King et al., 2018; Sloat et al., 2020). Several assessments have shown that results depend on the decision rule, the weighting scheme and the spatial support of the inputs (Bagwan & Gavali, 2024; Jelinski & Wu, 1996; Malone et al., 2015; Rashidi & Sharifian, 2022; Tenerelli & Carver, 2012; Xu & Zhang, 2013). Suitability is therefore not an intrinsic property of a land parcel but a product of the assessment specification (Snelder et al., 2023).

A specific and common specification problem arises when a threshold is transferred from a reference region where a system is already practised to a region where it is not. The threshold is then an empirical reference, and its value depends on how the reference is constructed: which climate source, period and temporal aggregation are used, whether the reference is a station point

or an area-weighted distribution over mapped cultivation, and which quantile of that distribution is taken. These choices are rarely reported as uncertainty, and the transferred map is rarely confronted with where the crop is actually mapped in the target region. Where it is, the comparison is usually reported as a single agreement figure at the chosen threshold, which cannot reveal whether the reference data could distinguish that threshold from plausible alternatives.

This study assesses that reliability problem for one concrete case: screening European arable land for the cold-region rice cultivation system of Hokkaido, Japan, the northernmost large-scale paddy rice system. I do not propose a new suitability method or a physiological threshold. Instead, I ask four assessment questions that can be answered with existing public monitoring and census-derived data. (i) How much does the construction of the empirical reference change the classification of a target region? (ii) How much of the mapped existing rice harvested area in Europe, and of the independently declared rice parcel area in France, is omitted by the thermal gate across a full range of thresholds? (iii) What is the trade-off between the area retained for next-stage assessment and the omitted existing rice area, and does that trade-off discriminate between competing references? (iv) How do the geographic denominator, the grid support and a non-compensatory irrigation-infrastructure gate change the interpretation? The contribution is empirical evidence on the instability and the limited verifiability of a transferred-threshold assessment, together with an explicit statement of what the available reference data can and cannot validate.

2. Data and methods

2.1 Domain, spatial unit and denominator

The assessment domain is Europe as delineated by Natural Earth 1:50 m country boundaries (Natural Earth, 2024), excluding Russia, whose territory is truncated by the 45°E edge of the 11°W–45°E, 34°N–71.5°N processing window. Transcontinental countries are assigned by the Natural Earth continent attribute, so that, for example, Türkiye, Georgia and Azerbaijan fall outside the European domain. All layers share one regular 5′ latitude–longitude grid aligned to the global origin (−180°, 90°). The denominator is the 2010 harvested area of wheat, maize, barley and potato in MapSPAM 2010 v2.0, all technologies combined (IFPRI, 2019; Yu et al., 2020), termed incumbent arable land. It totals 69.7 Mha in the domain and measures where arable cropping already occurs and where a new system would compete with incumbent crops; it is not an estimate of convertible land. Areas are harvested hectares as provided by MapSPAM; no additional pixel-area weighting is applied to them. Cell areas computed on the WGS84 authalic sphere are used only to express irrigation-equipped area as a share of each cell. The data sources and their roles are listed in Table 1.

Table 1 Data sources and their roles in the assessment.

Dataset	Version, support	Variable used	Role
WorldClim (Fick & Hijmans, 2017)	2.1; 1970–2000 normals; 2.5′, 5′, 10′	Monthly mean temperature	GDD10 target surface and grid reference calibration
MapSPAM (IFPRI, 2019; Yu et al., 2020)	2010 v2.0; 5′	Harvested area of wheat, maize, barley, potato; rice (all technologies)	Incumbent arable denominator; Hokkaido calibration; European rice coverage check
GMIA (Siebert et al., 2013)	v5; ~2005; 5′	Area equipped for irrigation	Non-compensatory infrastructure gate
NASA POWER (NASA POWER Project, 2025)	Daily point API v2; 1981–2024	Daily mean temperature at Hokkaido points	Point-derived alternative reference
RPG (IGN, 2026)	2010 (îlots), 2015, 2020, 2024; parcel polygons	Declared rice area (crop code RIZ, group 14)	Independent check of rice coverage in France
Natural Earth (Natural Earth, 2024)	Admin 0, 1:50 m, v5.1	Country polygons, continent attribute	European domain and country attribution

2.2 Thermal gate and reference construction

Commercial rice production in Hokkaido relies on ultra-early japonica cultivars, heated nurseries, transplanting and warm-water irrigation (MAFF, 2024). Hokkaido is used as the reference for this transferable cold-region system, not as a universal physiological limit. Thermal accumulation is GDD10, the sum of daily mean temperature above 10 °C over the season window of the original analysis pipeline, which runs from about 29 April to 11 September. On the grid, GDD10 is computed from WorldClim 2.1 monthly mean temperature for 1970–2000 (Fick & Hijmans, 2017), weighting each month by the number of its days inside the window. The primary grid reference T^* is calibrated on the same surface as the 5th percentile of GDD10 over Hokkaido rice cells, weighted by MapSPAM 2010 rice harvested area (114 thousand ha). The estimand of T^* is therefore the GDD10 exceeded by 95% of mapped Hokkaido rice harvested area on this specific climate surface, which is 589 GDD10; the 1st, 10th and 50th percentiles are 487, 637 and 859 GDD10.

The alternative reference used in an earlier version of the pipeline was a NASA POWER point series for 2000–2024 at 43.2°N, 143.0°E (NASA POWER Project, 2025; evaluated by Bai et al., 2010), which yields about 900 GDD10. To separate the effect of reference construction from that of spatial support, I recomputed the threshold along a controlled path on a fixed WorldClim 5′ target surface: point daily data 2000–2024, the same point for 1981–2000, a monthly-mean formula, the co-located WorldClim cell, the Hokkaido rice-area median and finally the rice-area 5th percentile. Grid support was assessed separately by recalibrating the 5th-percentile reference on 2.5′, 5′ and 10′ surfaces and applying each to its own surface. The same decomposition was repeated for a calendar June–September window.

2.3 Irrigation-infrastructure gate

A cell passes the infrastructure gate if its area equipped for irrigation in the Global Map of Irrigation Areas v5 (Siebert et al., 2013) is at least 10% of the cell area. This layer records equipped infrastructure around 2005 compiled from subnational statistics. It is not evidence of water allocation, rice-specific field infrastructure such as levelled basins and drainage, or implementation feasibility. The thermal and infrastructure gates are combined non-compensatorily: a cell is a candidate for next-stage assessment only if it passes both. I use "candidate" strictly in this sense; candidate area is not feasible or recommended land.

2.4 Threshold sweep against mapped existing rice area

The primary sweep holds the climate surface, season window, domain, denominator and weights fixed and varies only the threshold from 0 to 1800 GDD10 in steps of 25, adding the named references in Table 2. For each threshold T I compute the candidate share of incumbent arable land, weighted by incumbent harvested area a_i, and the omitted share of mapped rice harvested area, weighted by rice harvested area r_i, where G_i is the GDD10 of cell i:

$$C(T) = \frac{\sum_i a_i 1[G_i \geq T]}{\sum_i a_i} \qquad (1)$$

$$O(T) = \frac{\sum_i r_i 1[G_i < T]}{\sum_i r_i} \qquad (2)$$

Here $1[\cdot]$ is the indicator function and the sums run over domain cells with incumbent arable land (for C) or mapped rice area (for O).

Rice harvested area is taken from the same MapSPAM 2010 release (all technologies), covering 501.1 thousand ha in 20,356 domain cells. Omission is computed over all rice cells; the

variant restricted to cells that also contain incumbent arable land, used in earlier reports, is shown for comparison. Because the threshold is calibrated in Hokkaido and evaluated on European rice cells, the check is out of sample with respect to the calibration region. It is, however, a presence-only coverage check. Non-rice cropland is not an independently verified negative, because rice is excluded from most of Europe by water, soils, markets and institutions rather than by temperature alone. I therefore do not compute specificity, precision, F1 scores or receiver-operating-characteristic statistics, and the omission–candidate curve is not an ROC curve. On a fixed surface, $C(T)$ cannot increase and $O(T)$ cannot decrease with T; this monotonicity is mechanical. The information of the sweep lies in where along the threshold axis omissions begin to rise, relative to the competing references, and in how strongly candidate area changes over thresholds that the rice data cannot distinguish.

Table 2 Named threshold references and their consequences on the fixed primary surface (WorldClim 2.1 5′, 29 April–11 September). Cropland shares are weighted by incumbent arable harvested area; omission is weighted by mapped MapSPAM rice harvested area.

GDD10	Reference construction	Europe thermal pass (%)	Europe thermal + AEI (%)	Germany pass (%)	Rice omitted (%)
487	Hokkaido SPAM rice-area P01, WorldClim 5' (grid)	97.3	16.5	99.7	0.98
589	T*: Hokkaido SPAM rice-area P05, WorldClim 5' (grid; primary)	92.1	16.0	96.6	1.37
637	Hokkaido SPAM rice-area P10, WorldClim 5' (grid)	88.4	15.8	91.4	1.57
750	WorldClim 5' cell at the Hokkaido POWER point (grid point)	76.5	15.2	50.0	2.12
767	POWER Hokkaido point 1981-2000 monthly formula	74.1	15.0	41.5	2.19
832	POWER Hokkaido point 1981-2000 daily	63.1	14.7	14.7	2.64
859	Hokkaido SPAM rice-area P50, WorldClim 5' (grid)	59.4	14.5	9.0	2.93
900	~900: rounded POWER Hokkaido point 2000-2024 daily (earlier numerical reference)	54.9	14.2	3.2	3.22
903	POWER Hokkaido point 2000-2024 daily (unrounded)	54.6	14.2	3.1	3.23
1,036	All SPAM rice north of 35N, rice-area P05 (grid)	42.7	12.6	0.1	4.45
1,076	European SPAM rice-area P05 (in-sample for Europe; diagnostic only)	39.1	12.0	0.0	5.00

2.5 Omission geography, grid alignment and reference-data independence

Omitted rice area is mapped and summarised by country, by 8-connected cluster on the 5′ grid and by rice density within the cell. To test sensitivity to co-registration between MapSPAM and WorldClim, the rice layer was shifted by one cell in each of the eight directions, omission was recomputed with a 3 × 3 neighbourhood rule (a cell counts as omitted only if all neighbouring cells fail), with 2.5′ sub-cell pass fractions and on the native 10′ surface. Leave-one-country-out recomputation provides a range of omission shares that is a scenario range, not a statistical confidence interval; I do not bootstrap individual cells, because adjacent cells share climate and allocation information and are not independent observations.

MapSPAM harvested area is not an independent field observation. MapSPAM allocates national and subnational production statistics to grid cells using cropland extent, irrigation maps and crop-suitability information (Yu et al., 2020). Its rice layer is therefore partly conditioned on irrigation and climate information related to the gates evaluated here. The infrastructure gate shares the irrigation-statistics lineage of the MapSPAM irrigated technology, so agreement between mapped rice and the infrastructure gate is not treated as validation. For the thermal gate, the Hokkaido reference was calibrated on MapSPAM rice area in Japan, while the European rice area serves as the evaluation set; both originate from the same modelled product, so the check is out of region but not out of product.

2.6 Independent check against declared rice parcels in France

Because MapSPAM is not an independent observation, I repeated the omission check for France against the Registre parcellaire graphique (RPG), the anonymised register of agricultural parcels declared annually by farmers under the Common Agricultural Policy (IGN, 2026). The RPG crop code RIZ (crop group 14, Riz) is specific to rice. I used the 2015, 2020 and 2024

parcel layers and, for the reference year of MapSPAM, the 2010 layer, which records rice area by agricultural block (îlot) rather than by parcel. Only features coded as rice were retrieved from the official Géoplateforme web feature service, and overseas departments, which lie outside the European domain, were excluded. Parcel area was computed in the Lambert-93 projection, and each parcel was split across the 5′ climate cells it intersects in proportion to its area; assignment by parcel centroid is a sensitivity analysis. The estimand is the area-weighted share of declared rice area in cells that fail the thermal gate, evaluated over the same threshold sweep on the same primary surface. Parcels in one climate cell are not independent climate observations; the effective support is the number of distinct 5′ cells occupied, and I therefore report no parcel-level confidence intervals.

The RPG is an administrative register compiled for subsidy control, not a suitability model, and its geometry is not derived from climatic or irrigation layers. It may nevertheless contain declaration and digitising errors, and the subnational rice statistics that MapSPAM allocates may partly derive from the same declarations, so the check is independent of MapSPAM in spatial allocation but not necessarily in area totals. Like MapSPAM, it records presence only and cannot be used to estimate commission, specificity or classification accuracy.

2.7 Classification stability

Agreement between the primary specification and alternatives was computed on incumbent arable land: the point-derived 900 GDD10 reference, a June–September window recalibrated to its own 5th percentile (664 GDD10), 2.5′ and 10′ surfaces recalibrated to their own 5th percentiles (612 and 552 GDD10), and the Hokkaido rice-area median. For each pair I report the area-weighted agreement share and the Jaccard index of the pass sets, and I map the cells whose

class changes. Stability between specifications is reported separately from coverage of mapped rice area; neither establishes the agronomic validity of a gate.

2.8 Software and use of AI-assisted tools

All calculations were performed in Python with rasterio, geopandas, NumPy, SciPy and pandas. Every number in the text, tables and figures is generated by scripts from the cited data. An AI-assisted coding and writing agent (Devin, Cognition AI) was used to write and audit analysis scripts, reproduce calculations and draft text; I reviewed and edited all content and take full responsibility for it.

3. Results

3.1 Reference construction dominates the Germany contrast

In Germany (5.7 Mha of incumbent arable land) the pass share is 96.6% at T^* but 3.2% at the point-derived reference (Fig. 1). Along the controlled path on a fixed 5′ surface, changing the period from 2000–2024 to 1981–2000, replacing daily with monthly aggregation and moving from the point to its WorldClim cell raises the pass share to 50.0%; replacing the single cell with the Hokkaido rice-area distribution gives 9.0% at the median and 96.6% at the 5th percentile. In contrast, recalibrating the 5th-percentile reference at 2.5′, 5′ and 10′ yields thresholds of 612, 589 and 552 GDD10 and changes the German pass share by at most 2.4 percentage points. The quantile and the point-versus-distribution construction of the reference, not grid resolution, account for most of the contrast.

Fig. 1 Controlled decomposition of the German thermal pass share (incumbent arable land, 5.7 Mha). (a) Threshold recomputed along a path from the point-derived reference to the grid-calibrated T^* and applied to a fixed WorldClim 5′ surface. (b) Rice-area 5th-percentile reference recalibrated and applied at each resolution.

3.2 Thermal and infrastructure gates across Europe

At *T**, 92.1% of European incumbent arable land (64.2 Mha) passes the thermal gate, 54.9% at 900 GDD10 and 39.1% at 1,076 GDD10, the 5th percentile of mapped European rice area (Table 2). Adding the infrastructure gate reduces the candidate share at *T** to 16.0% (11.2 Mha), and to 14.2% at 900 GDD10. For most of the thermal range the infrastructure gate rather than the thermal gate determines the size of the candidate area, and no thermal surplus compensates for missing infrastructure.

3.3 Threshold-dependent omission of mapped rice area

Mapped European rice harvested area lies predominantly in warm cells: its rice-area-weighted GDD10 median is 1,413 and its 5th percentile is 1,076. Consequently the omission curve is nearly flat over the range of competing Hokkaido references (Fig. 2). Omission is 0.98% at the Hokkaido 1st percentile (487 GDD10), 1.37% at *T**, 2.93% at the Hokkaido median and 3.22% at 900 GDD10; it reaches 5.5% at 1100, 20.4% at 1300 and 45.2% at 1400 GDD10. Restricted to rice cells that also contain incumbent arable land, omission at *T** is 0.88%, the figure reported in earlier versions; the difference arises from 4.6 thousand ha of mapped rice in cells without incumbent crops.

Fig. 2 Threshold sweep on the fixed primary surface. Blue: share of incumbent arable land in Europe (cropland-area weighted) and Germany passing the thermal gate. Orange: share of mapped MapSPAM rice harvested area in Europe omitted by the gate (rice-area weighted). Dashed lines mark *T**, the point-derived reference and the 5th percentile of mapped European rice area.

3.4 Candidate area versus omitted rice area

The trade-off curve shows that the references proposed for Hokkaido differ greatly in candidate area but hardly in omission (Fig. 3). Moving from *T** to 900 GDD10 removes 37.2 percentage points (25.9 Mha) of incumbent arable land from the thermal pass set but adds only 1.9 percentage points of omitted rice area. In Germany the same move removes 93.3 percentage

points, while MapSPAM maps no rice in Germany. Omission first exceeds 10% of mapped rice area at 1,225 GDD10. The mapped rice data therefore reject thresholds that would exclude much of current Mediterranean and Danubian rice production but cannot discriminate among the Hokkaido-derived references that drive the European and German results.

Fig. 3 Trade-off between candidate incumbent arable area retained for next-stage assessment and omitted mapped rice harvested area across thresholds (0–1800 GDD10). Orange points are the named references in Table 2. Candidate area is not feasible land. This is not a receiver-operating-characteristic curve because non-rice cropland is not a verified negative.

3.5 Geography and robustness of omissions

At *T**, the 6,858 ha of omitted rice area fall in 485 cells forming 66 clusters; the five largest clusters contain 80% of the omitted area, and 85% of it lies in Italy and France (Fig. 4). In 96% of omitted cells, holding 64% of omitted area, rice occupies less than 1% of the cell, which is characteristic of statistical allocation to the cool upland margins of rice-producing administrative units, for example in northern Italy, south-eastern France and North Macedonia, rather than of established paddy districts. The omission share is sensitive to co-registration: one-cell shifts of the rice layer give 1.32–2.53% at *T**, and a 3 × 3 neighbourhood rule reduces it to 0.53% (Table 3). Leave-one-country-out recomputation gives 0.88–1.80% at *T** and 2.12–4.23% at 900 GDD10. These ranges are comparable in magnitude to the difference between *T** and 900 GDD10, reinforcing that the omission check cannot rank these references.

Fig. 4 Mapped MapSPAM 2010 rice harvested area in Europe by the GDD10 of its 5′ cell, showing where rice area is omitted at *T** = 589, 900 and 1100 GDD10. Marker area is proportional to rice harvested area (rice-area weights). ETRS89-LAEA projection.

Table 3 Sensitivity of the omitted share of mapped rice harvested area to co-registration and spatial support. Ranges are scenario ranges, not confidence intervals.

Test	**T* = 589**	**900**	**1100**
Rice layer shifted (+0,+0) cells	1.37	3.22	5.53
Rice layer shifted (-1,-1) cells	2.53	5.64	9.70
Rice layer shifted (+0,+1) cells	1.32	3.14	5.58
3 × 3 neighbourhood rule	0.53	1.70	3.19
2.5′ sub-cell pass fraction	1.33	3.23	5.78

Native 10′ surface	1.72	3.65	6.08
Leave-one-country-out range	0.88–1.80	2.12–4.23	4.01–7.15

3.6 Classification stability between specifications

Agreement with the primary specification is high for alternatives that keep the reference construction and vary support or window, and low for alternatives that change the reference itself (Table 4). The Jaccard index of the thermal pass sets is 0.98 for the recalibrated 2.5′ surface, 0.98 for the recalibrated 10′ surface and 0.93 for the recalibrated June–September window, but 0.60 for the 900 GDD10 reference and 0.64 for the Hokkaido median. In Germany the Jaccard index against the 900 GDD10 reference is 0.03. The disagreement between T^* and 900 GDD10 covers 25.9 Mha and forms a contiguous belt from western France through the Benelux countries, Germany, Poland and Belarus to the Baltic states (Fig. 5), which is the region in which transfer proposals for cold-tolerant rice would be most consequential. Of the candidate area that passes both gates at T^*, 1.2 Mha fails the thermal gate at 900 GDD10.

Fig. 5 Agreement of the thermal gate between T^* = 589 GDD10 and the point-derived 900 GDD10 on incumbent arable cells (cropland-area weights). No cell passes at 900 and fails at T^*, because the thresholds are nested on one surface. ETRS89-LAEA projection, 10 km display grid; classification on the native 5′ grid.

Table 4 Classification stability of the thermal gate relative to the primary specification (T^* = 589 GDD10, 5′, 29 April–11 September), weighted by incumbent arable harvested area.

Domain	Alternative specification	Agreement (%)	Jaccard (pass)	Pass in primary only (Mha)	Pass in alternative only (Mha)
Europe	900, 5', pipeline	62.8	0.60	25.94	0.00
Europe	P05 recalibrated Jun-Sep (664), 5'	93.4	0.93	4.63	0.00
Europe	P05 recalibrated 2.5' (612), majority of sub-cells	98.5	0.98	1.00	0.04
Europe	P05 recalibrated 10' (552)	97.8	0.98	0.09	1.48
Europe	Hokkaido P50 (859), 5'	67.2	0.64	22.83	0.00
Germany	900, 5', pipeline	6.7	0.03	5.31	0.00
Germany	P05 recalibrated Jun-Sep (664), 5'	89.2	0.89	0.62	0.00
Germany	P05 recalibrated 2.5' (612), majority of sub-cells	98.3	0.98	0.09	0.00
Germany	P05 recalibrated 10' (552)	97.9	0.98	0.01	0.11
Germany	Hokkaido P50 (859), 5'	12.4	0.09	4.98	0.00

3.7 Independent check against declared rice parcels in France

Declared rice area in metropolitan France is 13.7–19.6 thousand ha across the four RPG years (680 blocks in 2010 and 1,407–1,703 parcels in 2015–2024), similar to the 22.0 thousand ha that MapSPAM allocates to France. Its spatial support is very different (Fig. 6). Declared rice occupies only 29–39 distinct 5′ cells; at least 99% of its area lies south of 43.8°N and east of 4.2°E, in and around the Rhône delta (Camargue), and its area-weighted 5th percentile is 1,520 GDD10. MapSPAM spreads French rice over 865 cells, none holding more than 42 ha, with an area-weighted 5th percentile of 354 GDD10, and only 4.1% of its French rice area falls in the cells that contain declared rice in 2010. Consequently, the thermal gate omits 12.1% of MapSPAM rice area in France at T^*, 27.1% at 900 and 38.6% at 1100 GDD10, but at most 0.02% of declared rice area at any of these thresholds in any year; centroid assignment gives the same area shares and omits at most 0.24% of parcels by count. Within the cells that contain declared rice, MapSPAM omission is 0.0% at all three thresholds.

The omission of MapSPAM rice area in France is therefore an allocation effect outside the declared rice district, consistent with the low-density margin cells identified in Section 3.5, rather than evidence that the gate excludes current production. The independent records do not, however, help to rank references: every cell with declared rice exceeds 802 GDD10, and omission of declared rice area stays below 1% for every threshold up to 1,425 GDD10.

Fig. 6 Rice in metropolitan France on the primary GDD10 surface (WorldClim 2.1, 1970–2000, 29 April–11 September). (a) 5′ cells with MapSPAM 2010 rice harvested area; marker area increases with cell rice area. (b) 5′ cells with rice parcels declared in the RPG 2020; marker area increases with declared rice area in the cell. Blue and red lines are the 589 (T^*) and 900 GDD10 contours. (c) Share of rice area in cells below each threshold for MapSPAM 2010 in France and Europe and for declared rice area in RPG 2010 and 2020 (RPG 2015 and 2024 curves are indistinguishable from RPG 2020). Declared parcels are split across cells by area.

4. Discussion

4.1 Instability versus validity

Three results need to be kept apart. First, the classification of the target region is unstable with respect to reference construction: Germany changes from almost fully passing to almost fully failing depending on whether the reference is an area-weighted lower quantile on the target-consistent grid or a station-like point value. Second, the classification is stable with respect to grid support once the reference is recalibrated on each surface: scale effects of the modifiable areal unit problem (Jelinski & Wu, 1996) are present but small in this case, whereas the spatially explicit sensitivity to the reference itself (cf. Xu & Zhang, 2013) is large. Third, the available reference data on existing rice area validate neither specification in the critical range. Low omission at T^* demonstrates that the gate does not exclude current European rice production, but because European rice is mapped mainly in cells far warmer than any Hokkaido reference, omission would remain at or below 5% for every threshold up to 1,076 GDD10, which by construction is the 5th percentile of mapped European rice area. A permissive threshold can produce low omission mechanically, and the omission curve is informative only where it rises.

4.2 Implications for environmental suitability assessment

For transferred-threshold assessments, this case suggests four reporting practices. The estimand of the reference should be stated, including source, period, aggregation, spatial support and quantile, because these choices change results by an order of magnitude more than resolution does. The threshold should be swept, and the candidate area reported together with the omission of mapped existing cultivation, so that readers see which part of the threshold range the reference data can constrain. The independence of the reference data should be audited: crop maps produced by statistical allocation share inputs with the gates and are not field observations of

presence or absence. Where an administrative parcel register exists, as in France, it can separate allocation effects from omission of real production. Finally, non-compensatory gates such as irrigation infrastructure should be reported separately from climatic gates, because they can determine the candidate area regardless of the climatic reference. These practices complement multi-criteria uncertainty analysis (Malone et al., 2015; Tenerelli & Carver, 2012) and the recognition that suitability depends on the assessment frame (Snelder et al., 2023). They are relevant to suitability studies that rely on a few reference sites or on field-survey presence points (Benti et al., 2022; Wang et al., 2024).

4.3 Limitations and external validation

The analysis uses 1970–2000 climate normals and 2010 crop maps; it does not assess recent warming or observed land-use change. MapSPAM rice area is a modelled allocation of statistics, not mapped paddy fields, and harvested area may differ from physical area. GMIA records equipped infrastructure around 2005, not current water rights or rice-ready fields. The thermal gate represents one cold-region system; other cultivars, sowing dates or water-warming practices would require their own references. No independent negative data exist for rice absence, so commission cannot be estimated. The French parcel register checks coverage independently of MapSPAM but not the critical threshold range: declared rice lies in a few warm cells of one delta, the register years (2010–2024) postdate the 1970–2000 normals, so present-day GDD10 in these cells is probably higher, and the register records no failed attempts. A validation that could discriminate between references would require rice trials or cultivation records near the climatic margin, for example experimental plantings in central Europe, together with records of failed or abandoned attempts.

5. Conclusions

Transferring a thermal threshold from Hokkaido to Europe yields a candidate area that depends mainly on how the reference is constructed. Mapped existing rice area, and independently declared rice parcels in France, confirm that the gate does not exclude current European production, but neither can discriminate between references that differ by tens of millions of hectares of candidate land, and irrigation infrastructure further reduces the candidate area to 16.0% of incumbent arable land. Suitability maps based on transferred thresholds should therefore report reference construction, threshold sweeps and the limited verifiability of presence-only reference data as part of the assessment.

Declarations

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Competing interests

The author has no relevant financial or non-financial interests to disclose.

Ethics approval and consent

Not applicable; the study uses only publicly available gridded environmental and agricultural data.

Data availability

All input datasets are publicly available from the sources cited in Table 1 and the References. Derived outputs, including the threshold sweep, omission tables and classification-stability

tables, are available in the public repository https://github.com/bougtoir/more-rice. GMIA is not redistributed because of its licence, and RPG parcel geometries are not redistributed; derived cell-level RPG tables are included; the repository contains an acquisition ledger with source URLs, retrieval times, file sizes and SHA-256 checksums.

Code availability

All analysis and figure scripts are available in the public repository https://github.com/bougtoir/more-rice.

Author contributions

Tatsuki Onishi: conceptualization, methodology, software, formal analysis, data curation, visualization, writing – original draft, writing – review and editing.

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